# Supplementary material: every table generated from the saved runs

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All numbers are read from the run files by the code that builds this document. Cells are the mean over class partitions with the sample standard deviation as a subscript; brackets give a 95% $t$-interval over partitions.

| Score at 30% of the parameters | RESISC45 | | EuroSAT-MS | | PlantVillage | | PlantDoc | |
|---|---|---|---|---|---|---|---|---|
| | P | PMP | P | PMP | P | PMP | P | PMP |
| Dense ResNet-18, S0 only | $87.2_{\pm 2.5}$ | | $64.7_{\pm 4.9}$ | | $91.4_{\pm 5.6}$ | | $55.4_{\pm 1.3}$ | |
| Dense ResNet-18, S0+R+M+R | $87.0_{\pm 2.5}$ | | $64.2_{\pm 4.7}$ | | $91.3_{\pm 5.6}$ | | $55.5_{\pm 1.1}$ | |
| Random | $80.1_{\pm 3.6}$ | $81.1_{\pm 3.0}$ | $57.8_{\pm 6.7}$ | $58.4_{\pm 4.6}$ | $85.6_{\pm 8.1}$ | $86.8_{\pm 7.7}$ | $46.6_{\pm 1.3}$ | $46.7_{\pm 1.7}$ |
| Filter $\ell_1$ | $81.0_{\pm 3.1}^{\uparrow}$ | $81.5_{\pm 3.0}$ | $57.9_{\pm 6.2}$ | $59.4_{\pm 5.7}$ | $85.8_{\pm 8.3}$ | $86.4_{\pm 7.7}$ | $48.7_{\pm 0.9}^{\uparrow}$ | $48.1_{\pm 0.8}$ |
| BN scale $\lvert\gamma\rvert$ | – | $80.0_{\pm 3.1}^{\downarrow}$ | – | $56.8_{\pm 4.9}$ | $85.1_{\pm 8.9}$ | $85.1_{\pm 8.7}$ | $44.4_{\pm 1.7}^{\downarrow}$ | $45.5_{\pm 1.0}$ |
| Taylor | $82.5_{\pm 2.9}^{\uparrow}$ | $82.4_{\pm 3.0}^{\uparrow}$ | $60.2_{\pm 6.0}^{\uparrow}$ | $59.7_{\pm 6.3}$ | $88.3_{\pm 7.1}^{\uparrow}$ | $87.1_{\pm 7.1}$ | $47.9_{\pm 1.2}$ | $48.5_{\pm 1.0}^{\uparrow}$ |
| Gradient $\mathcal{G}$ | – | $81.4_{\pm 3.3}$ | – | $58.9_{\pm 5.6}$ | $87.3_{\pm 7.2}$ | $86.7_{\pm 7.9}$ | $47.6_{\pm 1.5}$ | $46.9_{\pm 2.5}$ |
| Variance $\mathcal{V}$ | – | $81.5_{\pm 3.0}$ | – | $57.3_{\pm 4.4}$ | $86.5_{\pm 6.9}$ | $86.5_{\pm 7.3}$ | $47.2_{\pm 1.4}$ | $47.3_{\pm 1.3}$ |
| Fisher $\mathcal{D}$ | $79.6_{\pm 3.1}$ | $79.9_{\pm 2.9}^{\downarrow}$ | $58.1_{\pm 6.5}$ | $58.4_{\pm 6.0}$ | $85.2_{\pm 8.9}$ | $85.3_{\pm 8.0}$ | $46.8_{\pm 0.8}$ | $46.4_{\pm 1.3}$ |
| DACIS | $80.1_{\pm 3.0}$ | $81.0_{\pm 3.3}$ | $57.8_{\pm 5.3}$ | $59.3_{\pm 4.9}$ | $86.4_{\pm 7.7}$ | $86.6_{\pm 7.5}$ | $47.3_{\pm 1.4}$ | $47.5_{\pm 1.6}$ |

Table 1. All eight scores at 30% of the parameters, schedules P and PMP.

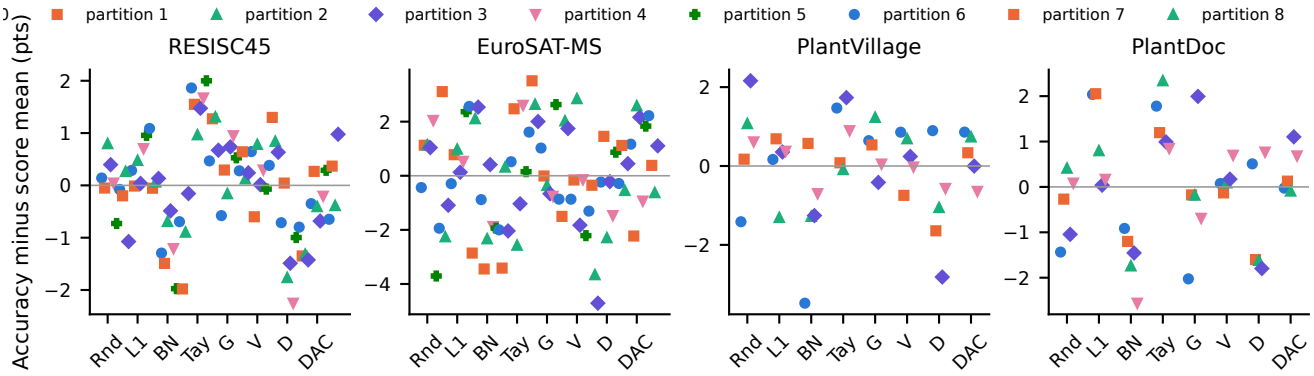



Figure 1. 5-shot accuracy of each score at 30% of the parameters (schedule PMP) on each class partition, relative to the mean over the eight scores; markers are partitions, offset horizontally for legibility. Rnd: random, L1: filter $\ell_1$, BN: BatchNorm scale, Tay: Taylor, G, V, D: gradient, variance, Fisher, DAC: DACIS.

| Schedule | Steps after S0 |
|---|---|
| P | P(30%), R |
| P→M | P(30%), R, M |
| M→P | M, P(30%), R |
| P→G→P | P(60%), R, G, P(30%), R |
| PMP | P(60%), R, M, P(30%), R |
| PMPM | PMP, then M |
| P→M, 2× | P(30%), R, M with twice the episodes |

Table 2. The seven schedules (P(x): prune to share x of the parameters; G: gradient probe).

| Difference (a − b), points | RESISC45 | EuroSAT-MS | PlantVillage | PlantDoc |
|---|---|---|---|---|
| *Scores, schedule P* | | | | |
| DACIS − Random | −0.1 [−0.9,+0.8] | +0.1 [−2.6,+2.8] | +0.8 [−1.3,+3.0] | +0.7 [−1.2,+2.6] |
| DACIS − Filter $\ell_1$ | −0.9 [−1.5,−0.3] | −0.1 [−2.2,+2.1] | +0.6 [−1.9,+3.1] | −1.4 [−3.3,+0.5] |
| DACIS − Taylor | −2.4 [−3.0,−1.7] | −2.4 [−4.9,+0.1] | −1.9 [−2.9,−0.9] | −0.6 [−1.6,+0.4] |
| *Scores, schedule PMP* | | | | |
| DACIS − Random | 0.0 [−0.8,+0.7] | +0.9 [−1.3,+3.0] | −0.3 [−2.3,+1.8] | +0.8 [−0.4,+2.1] |
| DACIS − Filter $\ell_1$ | −0.4 [−1.0,+0.1] | −0.1 [−1.6,+1.4] | +0.2 [−1.3,+1.7] | −0.7 [−2.4,+1.1] |
| DACIS − Taylor | −1.4 [−1.9,−0.9] | −0.4 [−2.7,+1.9] | −0.6 [−1.9,+0.8] | −1.1 [−2.4,+0.3] |
| *Episodic stage on the dense network* | | | | |
| Dense+M − Dense | −0.3 [−0.4,−0.1] | −0.8 [−1.5,0.0] | −0.3 [−0.6,+0.1] | 0.0 [−0.4,+0.3] |

Table 3. Paired differences between choices, in points.

| Difference (a − b), points | RESISC45 | EuroSAT-MS | PlantVillage | PlantDoc |
|---|---|---|---|---|
| *Schedules, DACIS at 30%* | | | | |
| PMP − P | +0.9 [+0.4,+1.5] | +1.4 [−0.3,+3.2] | +0.1 [−0.7,+0.9] | +0.1 [−0.8,+1.1] |
| P→M − P | 0.0 [−0.6,+0.5] | +0.7 [−0.9,+2.3] | −0.7 [−2.4,+1.0] | −0.8 [−1.6,0.0] |
| M→P − P | +0.9 [+0.3,+1.4] | +0.9 [−0.7,+2.4] | +0.3 [−1.2,+1.7] | −0.7 [−1.5,+0.2] |
| P→G→P (no weight update) − P | +1.2 [+0.6,+1.8] | +0.7 [−0.6,+2.0] | +0.3 [−0.4,+1.0] | −0.2 [−0.8,+0.5] |
| P→M→P→M − PMP | – | – | −0.1 [−1.2,+1.1] | +0.1 [−1.0,+1.2] |
| P→M (2× episodes) − P→M | – | – | −0.1 [−0.7,+0.4] | 0.0 [−1.0,+0.9] |

Table 4. Paired differences between schedules for DACIS at 30% of the parameters, in points.

| | RESISC45 | | | EuroSAT-MS | | | PlantVillage | | |
|---|---|---|---|---|---|---|---|---|---|
| Model | 1-shot | 5-shot | 10-shot | 1-shot | 5-shot | 10-shot | 1-shot | 5-shot | 10-shot |
| Dense ResNet-18 | $71.1_{\pm 3.6}$ | $87.2_{\pm 2.5}$ | $89.5_{\pm 2.3}$ | $51.3_{\pm 5.3}$ | $64.7_{\pm 4.9}$ | $67.6_{\pm 4.8}$ | $79.5_{\pm 9.6}$ | $91.4_{\pm 5.6}$ | $93.1_{\pm 4.4}$ |
| Random, PMP 30% | $64.6_{\pm 3.6}$ | $81.1_{\pm 3.0}$ | $84.0_{\pm 2.8}$ | $46.6_{\pm 5.1}$ | $58.4_{\pm 4.6}$ | $61.4_{\pm 4.5}$ | $73.6_{\pm 10.9}$ | $86.8_{\pm 7.7}$ | $88.9_{\pm 6.5}$ |
| Filter $\ell_1$, PMP 30% | $65.6_{\pm 3.5}$ | $81.5_{\pm 3.0}$ | $84.3_{\pm 2.8}$ | $47.8_{\pm 6.2}$ | $59.4_{\pm 5.7}$ | $62.0_{\pm 5.5}$ | $73.5_{\pm 11.1}$ | $86.4_{\pm 7.7}$ | $88.7_{\pm 6.4}$ |
| Taylor, PMP 30% | $64.7_{\pm 3.8}$ | $82.4_{\pm 3.0}$ | $85.1_{\pm 2.8}$ | $47.7_{\pm 6.5}$ | $59.7_{\pm 6.3}$ | $62.5_{\pm 6.0}$ | $72.9_{\pm 11.0}$ | $87.1_{\pm 7.1}$ | $89.4_{\pm 5.8}$ |
| DACIS, PMP 30% | $63.5_{\pm 3.9}$ | $81.0_{\pm 3.3}$ | $84.0_{\pm 2.9}$ | $47.7_{\pm 5.8}$ | $59.3_{\pm 4.9}$ | $61.9_{\pm 4.7}$ | $72.7_{\pm 10.8}$ | $86.6_{\pm 7.5}$ | $88.9_{\pm 6.3}$ |
| MobileNetV3-Small | $73.1_{\pm 3.5}$ | $88.1_{\pm 2.5}$ | $90.3_{\pm 2.2}$ | $60.2_{\pm 8.3}$ | $74.7_{\pm 6.0}$ | $78.0_{\pm 5.3}$ | $78.0_{\pm 10.1}$ | $91.1_{\pm 5.9}$ | $93.1_{\pm 4.7}$ |

Table 5. 1-, 5- and 10-shot accuracy (%) of the main models on the three novel-class sets.

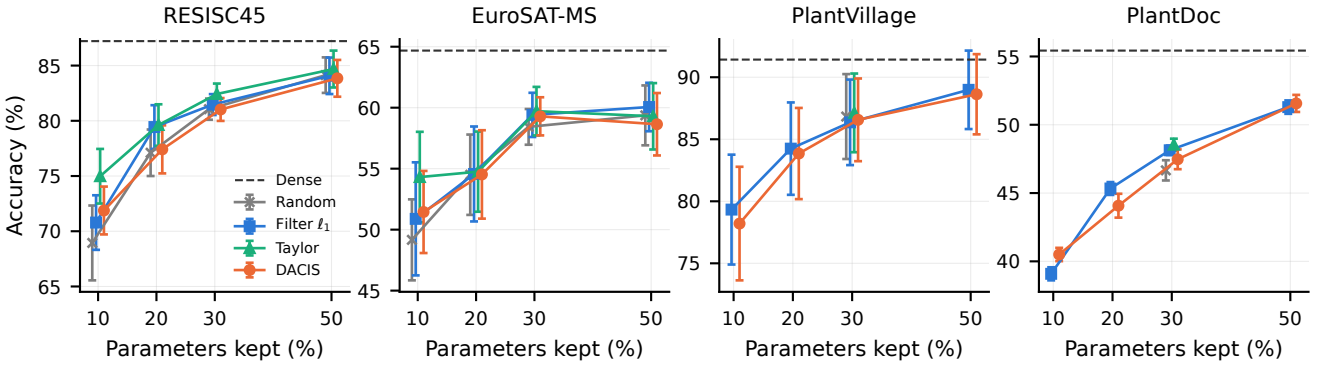



Figure 2. 5-shot accuracy against the share of parameters kept for four scores (schedule PMP); error bars are the standard error over partitions; dashed: dense network.

| Schedule (DACIS, 30%), 5-shot | RESISC45 | EuroSAT-MS | PlantVillage | PlantDoc |
|---|---|---|---|---|
| P | $80.1_{\pm 3.0}$ | $57.8_{\pm 5.3}$ | $86.4_{\pm 7.7}$ | $47.3_{\pm 1.4}$ |
| P→M | $80.0_{\pm 3.3}$ | $58.5_{\pm 5.9}$ | $85.8_{\pm 7.6}$ | $46.5_{\pm 1.3}$ |
| P→M (2×) | – | – | $85.6_{\pm 7.3}$ | $46.4_{\pm 1.2}$ |
| M→P | $80.9_{\pm 3.2}$ | $58.7_{\pm 5.4}$ | $86.7_{\pm 6.9}$ | $46.7_{\pm 0.9}$ |
| P→G→P | $81.3_{\pm 3.2}$ | $58.6_{\pm 5.4}$ | $86.8_{\pm 7.4}$ | $47.1_{\pm 1.4}$ |
| P→M→P (PMP) | $81.0_{\pm 3.3}$ | $59.3_{\pm 4.9}$ | $86.6_{\pm 7.5}$ | $47.5_{\pm 1.6}$ |
| P→M→P→M | – | – | $86.5_{\pm 7.4}$ | $47.6_{\pm 1.6}$ |

Table 6. Schedules for DACIS at 30% of the parameters, 5-shot accuracy.

| Params kept | Random | Filter $\ell_1$ | Taylor | DACIS |
|---|---|---|---|---|
| *RESISC45* | | | | |
| 50% | $84.1_{\pm 3.6}$ | $84.1_{\pm 3.7}$ | $84.7_{\pm 3.7}$ | $83.8_{\pm 3.7}$ |
| 30% | $81.1_{\pm 3.0}$ | $81.5_{\pm 3.0}$ | $82.4_{\pm 3.0}$ | $81.0_{\pm 3.3}$ |
| 20% | $77.1_{\pm 4.7}$ | $79.4_{\pm 4.4}$ | $79.6_{\pm 4.2}$ | $77.4_{\pm 4.9}$ |
| 10% | $68.9_{\pm 7.6}$ | $70.8_{\pm 5.5}$ | $75.0_{\pm 5.5}$ | $71.9_{\pm 4.8}$ |
| *EuroSAT-MS* | | | | |
| 50% | $59.4_{\pm 5.5}$ | $60.1_{\pm 4.4}$ | $59.3_{\pm 6.1}$ | $58.6_{\pm 5.7}$ |
| 30% | $58.4_{\pm 4.6}$ | $59.4_{\pm 5.7}$ | $59.7_{\pm 6.3}$ | $59.3_{\pm 4.9}$ |
| 20% | $54.5_{\pm 7.4}$ | $54.6_{\pm 8.7}$ | $54.8_{\pm 7.3}$ | $54.5_{\pm 8.1}$ |
| 10% | $49.2_{\pm 7.4}$ | $50.9_{\pm 10.4}$ | $54.3_{\pm 8.3}$ | $51.4_{\pm 7.5}$ |
| *PlantVillage* | | | | |
| 50% | – | $89.0_{\pm 7.1}$ | – | $88.6_{\pm 7.2}$ |
| 30% | $86.8_{\pm 7.7}$ | $86.4_{\pm 7.7}$ | $87.1_{\pm 7.1}$ | $86.6_{\pm 7.5}$ |
| 20% | – | $84.2_{\pm 8.3}$ | – | $83.8_{\pm 8.2}$ |
| 10% | – | $79.3_{\pm 9.9}$ | – | $78.2_{\pm 10.2}$ |
| *PlantDoc* | | | | |
| 50% | – | $51.3_{\pm 1.1}$ | – | $51.6_{\pm 1.4}$ |
| 30% | $46.7_{\pm 1.7}$ | $48.1_{\pm 0.8}$ | $48.5_{\pm 1.0}$ | $47.5_{\pm 1.6}$ |
| 20% | – | $45.3_{\pm 1.0}$ | – | $44.1_{\pm 2.0}$ |
| 10% | – | $39.1_{\pm 1.1}$ | – | $40.5_{\pm 1.1}$ |

Table 7. 5-shot accuracy of four scores at four budgets (schedule PMP), one block per test setting.

| PlantVillage 5-shot, schedule P, 30% | 1 epoch | 3 epochs | 6 epochs |
|---|---|---|---|
| Random | $85.6_{\pm 8.1}$ | $86.8_{\pm 8.1}$ | $86.3_{\pm 8.2}$ |
| Filter $\ell_1$ | $85.8_{\pm 8.3}$ | $87.4_{\pm 7.3}$ | $87.3_{\pm 7.2}$ |
| Taylor | $88.3_{\pm 7.1}$ | $88.2_{\pm 7.3}$ | $87.8_{\pm 7.7}$ |
| DACIS | $86.4_{\pm 7.7}$ | $86.6_{\pm 7.7}$ | $86.5_{\pm 7.7}$ |
| Dense, no pruning | | $91.4_{\pm 5.6}$ | |

Table 8. Recovery length after pruning (PlantVillage).

| DACIS variant (PMP, 30%) | PV 5-shot | Δ vs. full |
|---|---|---|
| DACIS (full) | $86.6_{\pm 7.5}$ | – |
| $\mathcal{G}$ only | $86.7_{\pm 7.9}$ | +0.2 [−0.4,+0.7] |
| $\mathcal{V}$ only | $86.5_{\pm 7.3}$ | −0.1 [−0.8,+0.7] |
| $\mathcal{D}$ only | $85.3_{\pm 8.0}$ | −1.3 [−2.9,+0.3] |
| $\mathcal{G}$+$\mathcal{V}$ | $86.9_{\pm 7.0}$ | +0.4 [−0.7,+1.5] |
| $\mathcal{G}$+$\mathcal{D}$ | $86.5_{\pm 7.5}$ | −0.1 [−1.2,+1.1] |
| $\mathcal{V}$+$\mathcal{D}$ | $86.2_{\pm 7.6}$ | −0.4 [−1.5,+0.7] |
| no meta-gradient term | $85.6_{\pm 7.6}$ | −0.9 [−1.7,−0.1] |
| same width at every depth | $86.1_{\pm 7.6}$ | −0.5 [−1.2,+0.1] |
| min-max instead of ranks | $86.1_{\pm 7.9}$ | −0.5 [−1.2,+0.2] |

Table 9. DACIS components and design choices (PlantVillage).

| $(\lambda_G, \lambda_V, \lambda_D)$ | Val 1-shot | Val 5-shot | Novel 1-shot | Novel 5-shot |
|---|---|---|---|---|
| (0.3,0.2,0.5) | $74.3_{\pm 6.8}$ | $88.5_{\pm 4.2}$ | $73.3_{\pm 10.4}$ | $86.9_{\pm 7.2}$ |
| (0.33,0.33,0.33) | $73.7_{\pm 6.2}$ | $88.1_{\pm 4.0}$ | $73.2_{\pm 10.7}$ | $86.9_{\pm 7.5}$ |
| (0.5,0.2,0.3) | $73.6_{\pm 6.5}$ | $87.9_{\pm 4.5}$ | $72.3_{\pm 10.8}$ | $86.2_{\pm 7.7}$ |
| (0.2,0.5,0.3) | $74.3_{\pm 6.0}$ | $88.5_{\pm 4.0}$ | $72.7_{\pm 10.9}$ | $86.5_{\pm 7.3}$ |
| (0.2,0.2,0.6) | $74.3_{\pm 6.9}$ | $88.9_{\pm 4.0}$ | $72.4_{\pm 10.7}$ | $86.2_{\pm 7.8}$ |
| (0.1,0.1,0.8) | $74.3_{\pm 6.3}$ | $88.2_{\pm 4.2}$ | $72.3_{\pm 10.9}$ | $86.1_{\pm 7.8}$ |
| (0,0,1) | $74.1_{\pm 7.4}$ | $88.0_{\pm 4.6}$ | $71.8_{\pm 10.2}$ | $85.9_{\pm 7.8}$ |
| (0.5,0.5,0) | $74.1_{\pm 5.7}$ | $88.8_{\pm 3.4}$ | $73.0_{\pm 10.8}$ | $86.9_{\pm 7.0}$ |

Table 10. Weights of DACIS on validation and novel classes (PlantVillage).

| Episodic learning rate | PlantVillage val. | | RESISC45 val. | |
|---|---|---|---|---|
| | 1-shot | 5-shot | 1-shot | 5-shot |
| none (S0 only) | 86.7 | 96.1 | 74.2 | 90.1 |
| 0.0001 | 85.4 | 95.3 | 75.8 | 90.0 |
| 0.00025 | 84.9 | 95.0 | 76.1 | 89.8 |
| 0.0005 | 84.3 | 94.7 | 75.9 | 89.5 |
| 0.001 | 83.6 | 94.5 | 75.2 | 89.0 |
| 0.002 | 82.7 | 94.2 | 75.0 | 89.1 |
| 1e-05 (extra check) | – | – | 74.5 | 90.1 |
| 2.5e-05 (extra check) | – | – | 74.9 | 90.1 |
| 5e-05 (extra check) | – | – | 75.3 | 90.1 |

Table 11. Learning rate of the episodic stage, 1- and 5-shot accuracy on the validation classes of the first partition. The PlantVillage column chose the value used everywhere; the RESISC45 sweep is a check.

| EuroSAT-MS, novel classes | 1-shot | 5-shot | 10-shot |
|---|---|---|---|
| 13 bands, dense (S0) | $51.3_{\pm 5.3}$ | $64.7_{\pm 4.9}$ | $67.6_{\pm 4.8}$ |
| RGB bands, dense (S0) | $53.7_{\pm 4.3}$ | $68.7_{\pm 3.9}$ | $71.4_{\pm 3.6}$ |
| 13 bands, dense (S0+R+M+R) | $50.8_{\pm 5.0}$ | $64.2_{\pm 4.7}$ | $67.0_{\pm 4.7}$ |
| RGB bands, dense (S0+R+M+R) | $53.1_{\pm 4.5}$ | $68.1_{\pm 3.8}$ | $70.9_{\pm 3.4}$ |
| 13 bands, filter $\ell_1$, PMP 30% | $47.8_{\pm 6.2}$ | $59.4_{\pm 5.7}$ | $62.0_{\pm 5.5}$ |
| RGB bands, filter $\ell_1$, PMP 30% | $49.8_{\pm 2.4}$ | $63.9_{\pm 2.6}$ | $66.6_{\pm 2.3}$ |
| 13 bands, DACIS, PMP 30% | $47.7_{\pm 5.8}$ | $59.3_{\pm 4.9}$ | $61.9_{\pm 4.7}$ |
| RGB bands, DACIS, PMP 30% | $50.2_{\pm 3.5}$ | $63.6_{\pm 3.9}$ | $66.2_{\pm 3.7}$ |
| 13 bands, MobileNetV3-Small | $60.2_{\pm 8.3}$ | $74.7_{\pm 6.0}$ | $78.0_{\pm 5.3}$ |
| RGB bands, MobileNetV3-Small | $58.3_{\pm 3.7}$ | $74.7_{\pm 4.1}$ | $78.0_{\pm 4.1}$ |
| 13 bands − RGB, dense (5-shot) | | −6.6 [−17.4,+4.2] | |
| 13 bands − RGB, DACIS PMP (5-shot) | | −6.4 [−15.5,+2.6] | |

Table 12. EuroSAT-MS with 13 bands against the three visible bands.

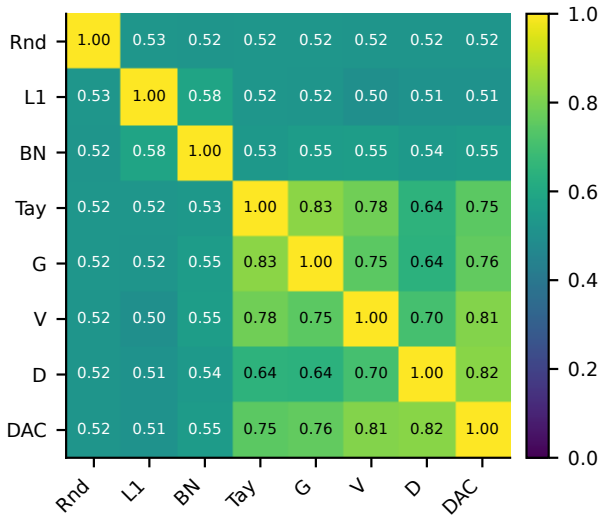



Figure 3. Overlap (Jaccard) of the channels kept by the first pruning step of different scores (RESISC45 if available, else PlantVillage).

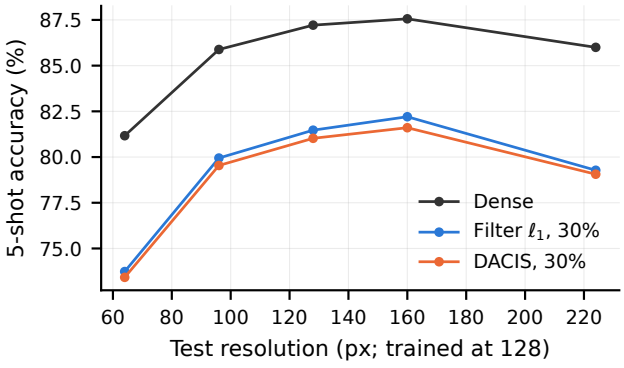



Figure 4. 5-shot accuracy when the test resolution differs from the training resolution (RESISC45 if available, else PlantVillage).